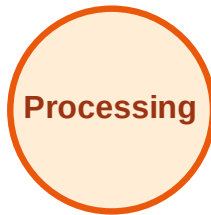


BFS Traversal

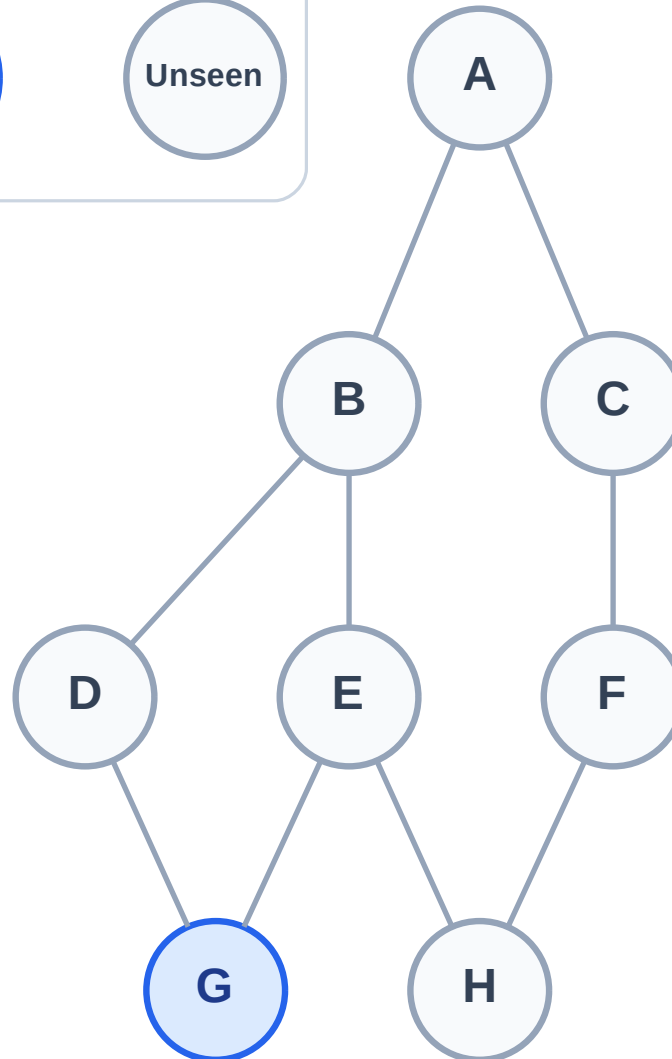
Step 1: Start at G

Legend



Queue: G

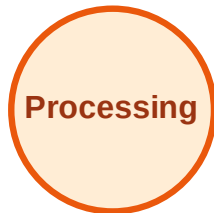
Visited: (none)



BFS Traversal

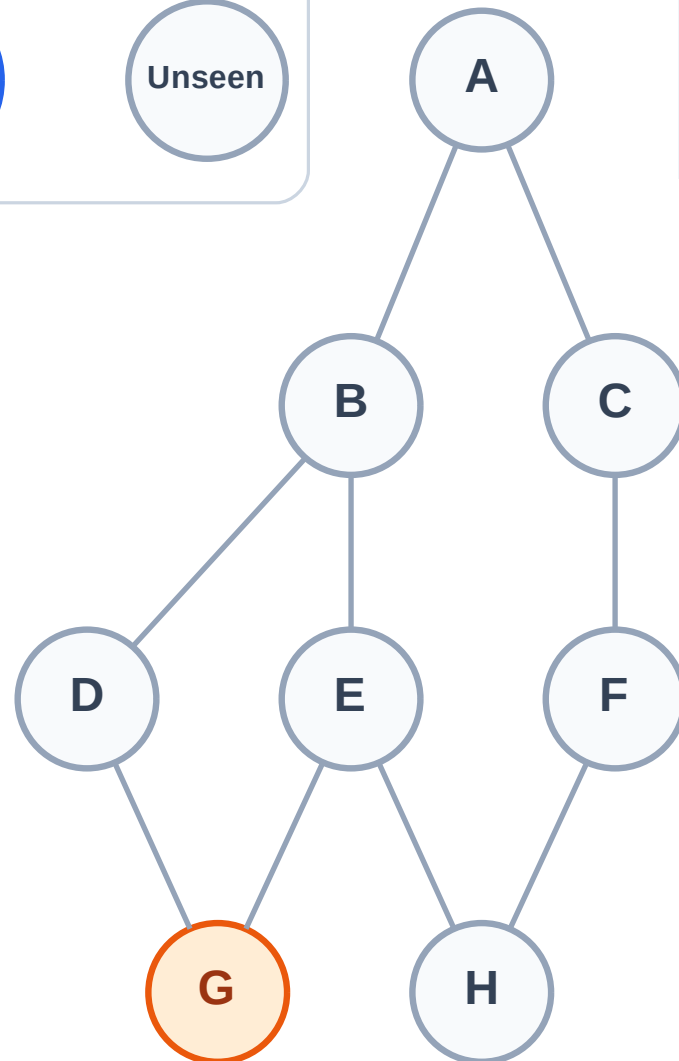
Step 2: Process node G

Legend



Queue: (empty)

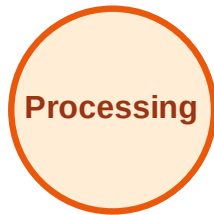
Visited: (none)



BFS Traversal

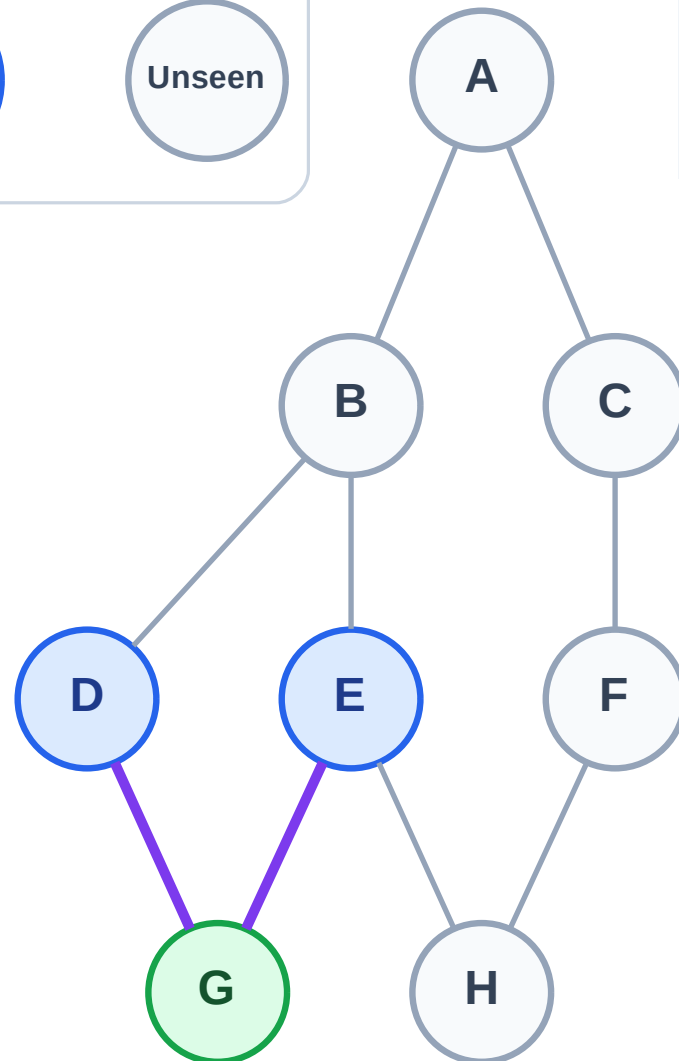
Step 3: Discover D, E from G

Legend



Queue: D → E

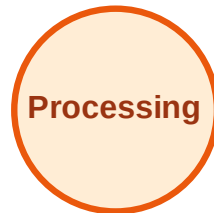
Visited: G



BFS Traversal

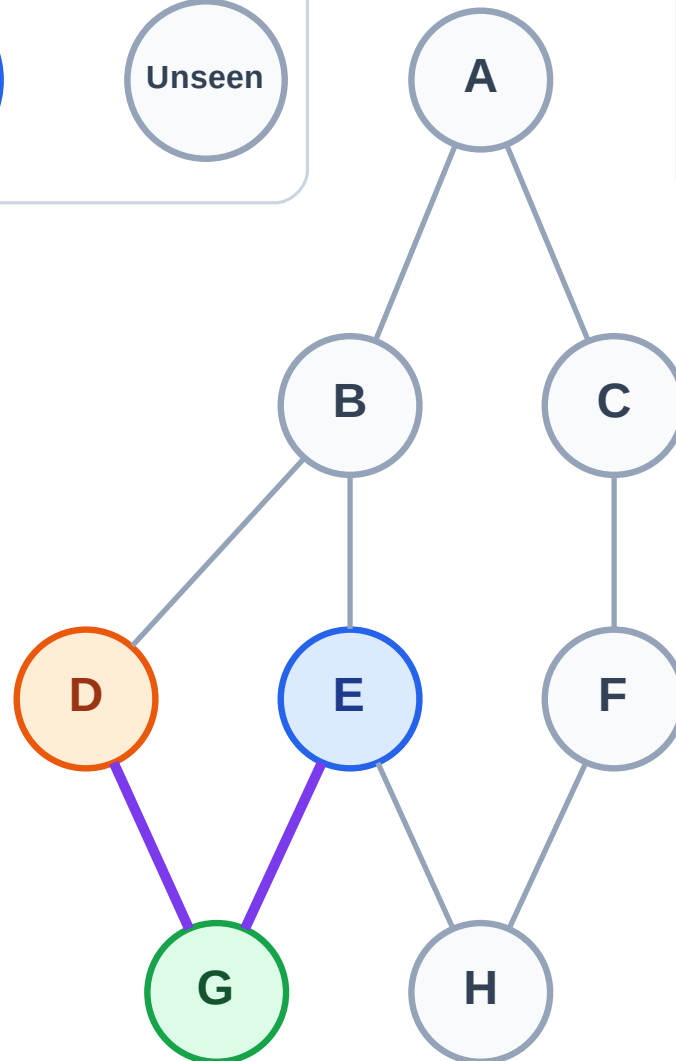
Step 4: Process node D

Legend



Queue: E

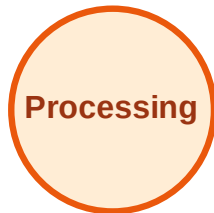
Visited: G



BFS Traversal

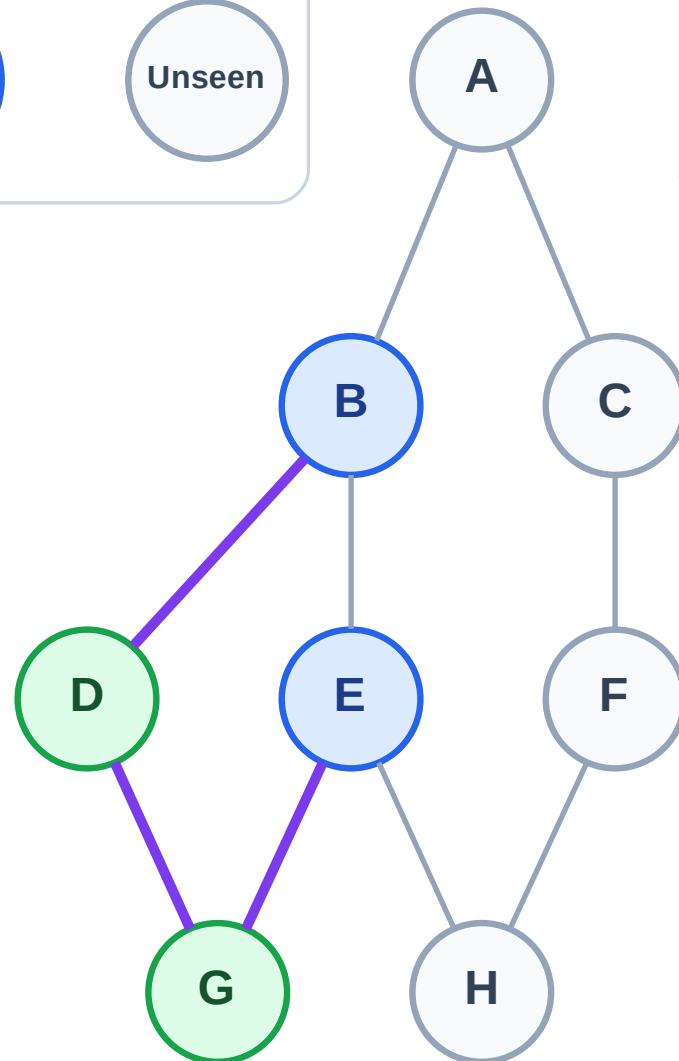
Step 5: Discover B from D

Legend



Queue: E → B

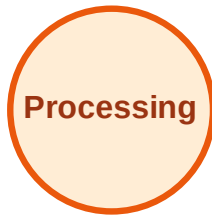
Visited: D, G



BFS Traversal

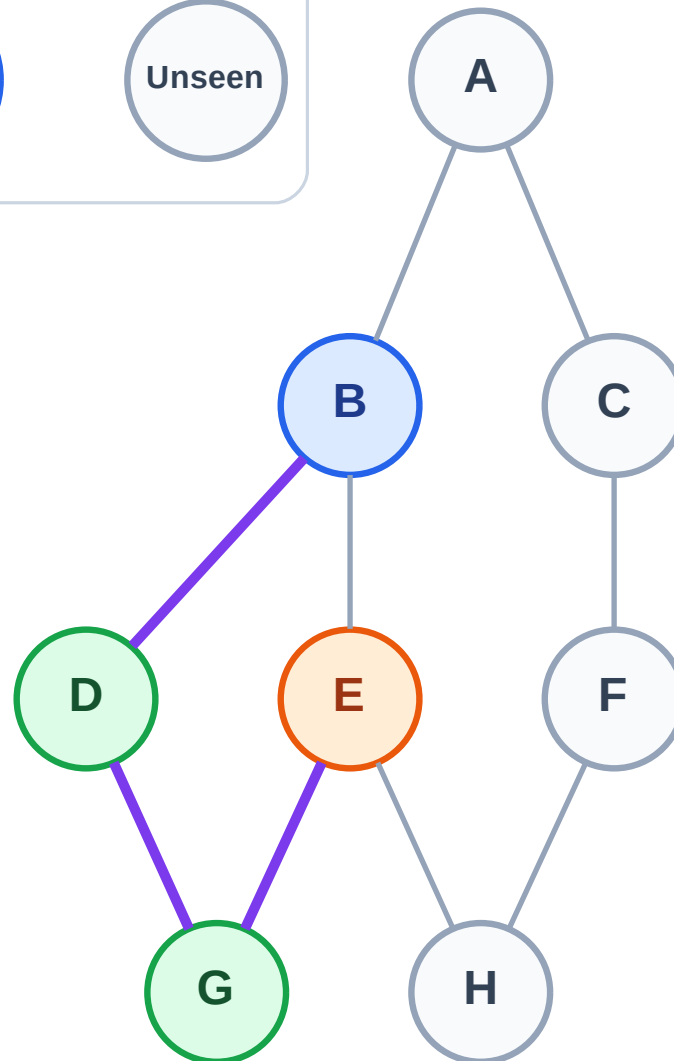
Step 6: Process node E

Legend



Queue: B

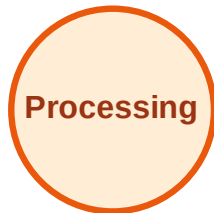
Visited: D, G



BFS Traversal

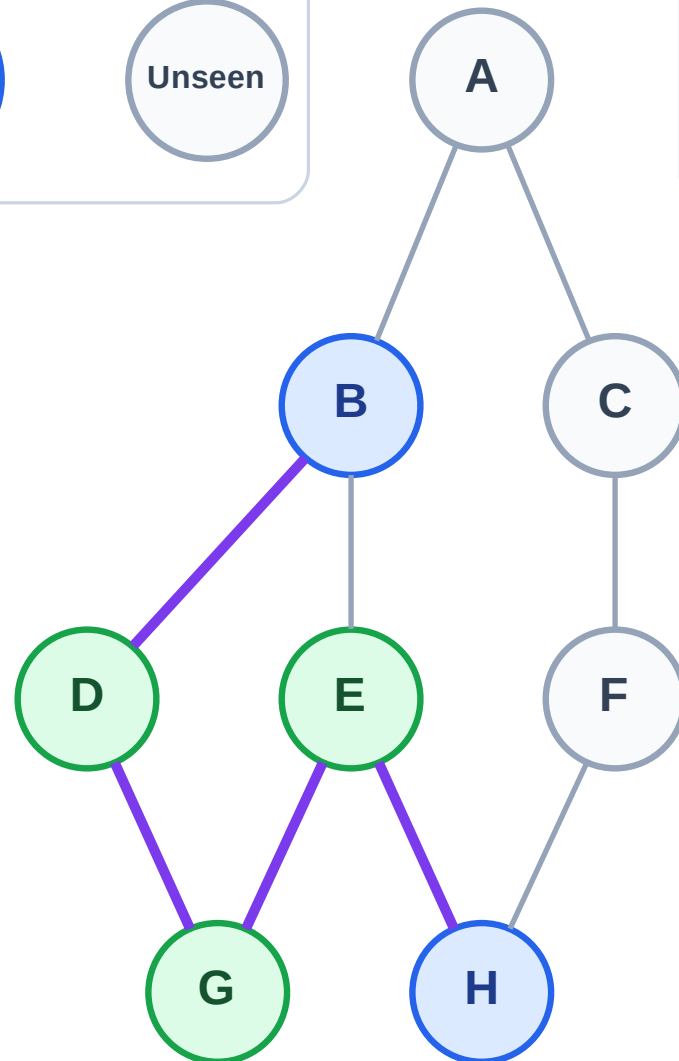
Step 7: Discover H from E

Legend



Queue: B → H

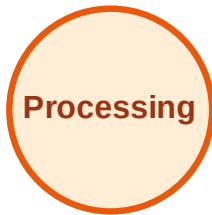
Visited: D, E, G



BFS Traversal

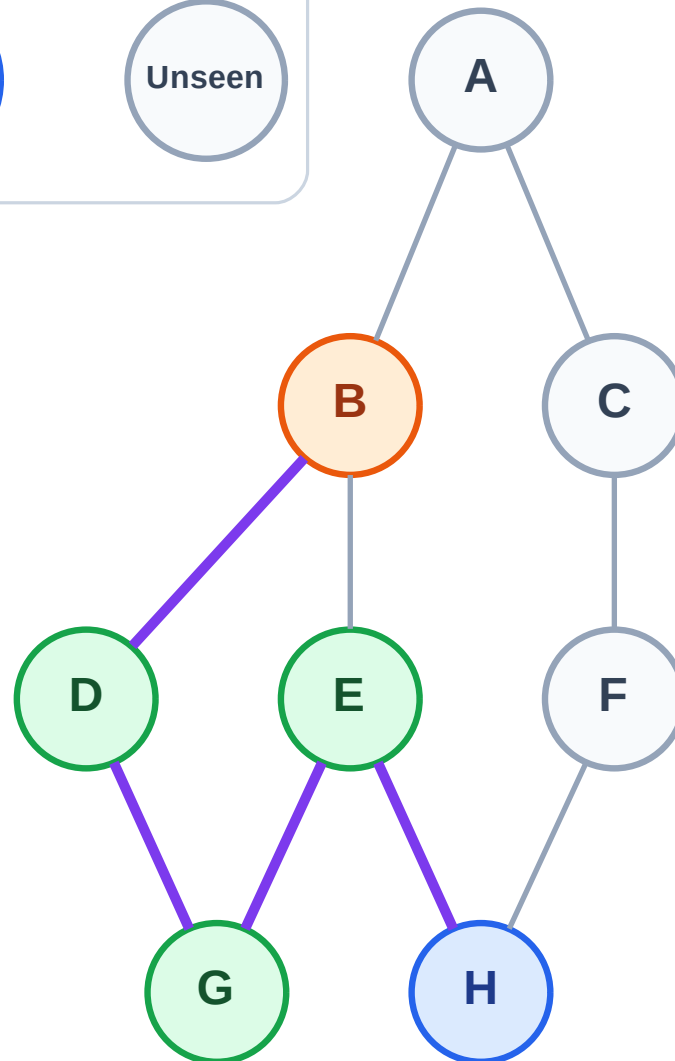
Step 8: Process node B

Legend



Queue: H

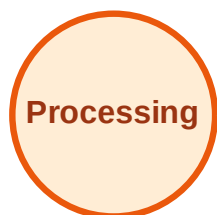
Visited: D, E, G



BFS Traversal

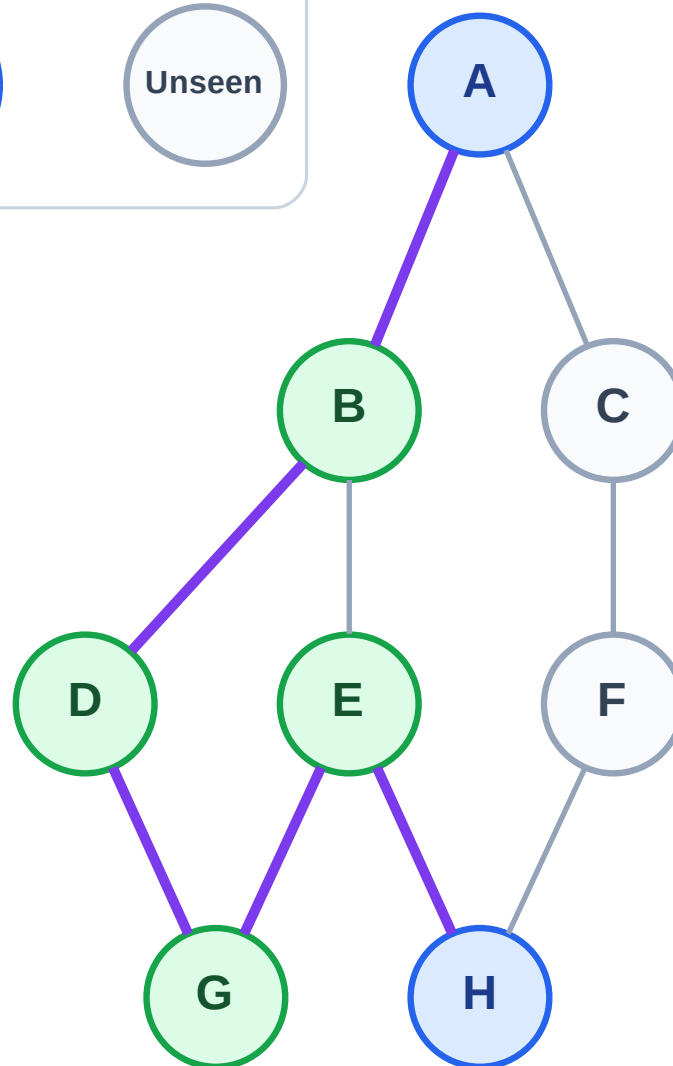
Step 9: Discover A from B

Legend



Queue: H → A

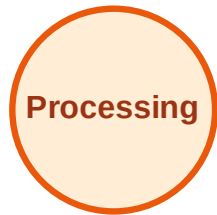
Visited: B, D, E, G



BFS Traversal

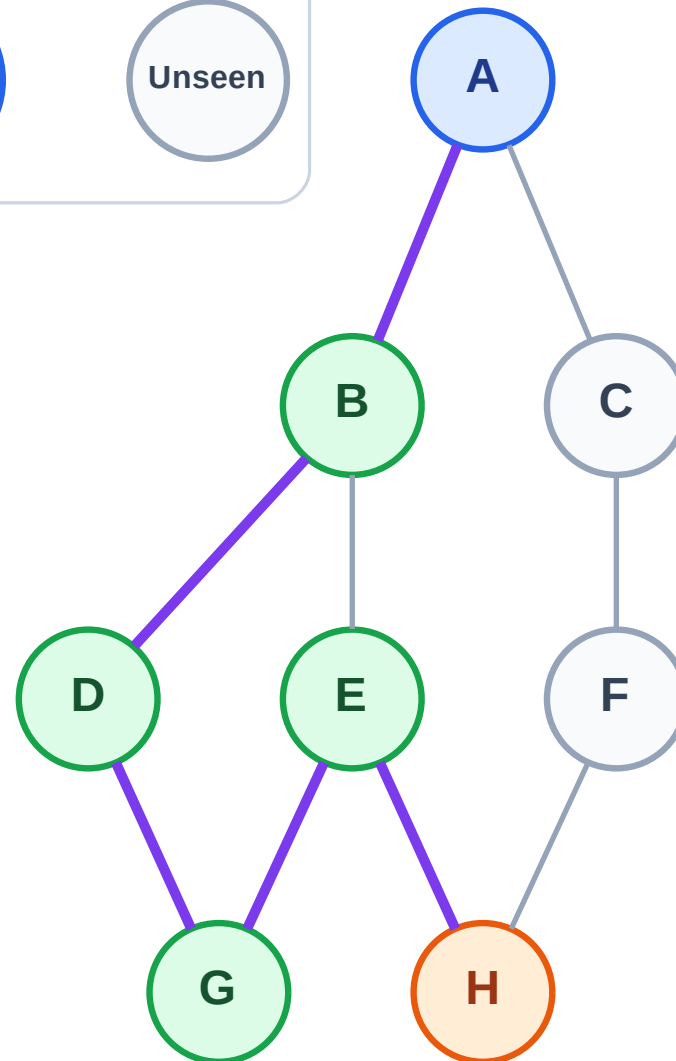
Step 10: Process node H

Legend



Queue: A

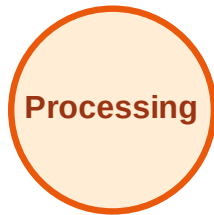
Visited: B, D, E, G



BFS Traversal

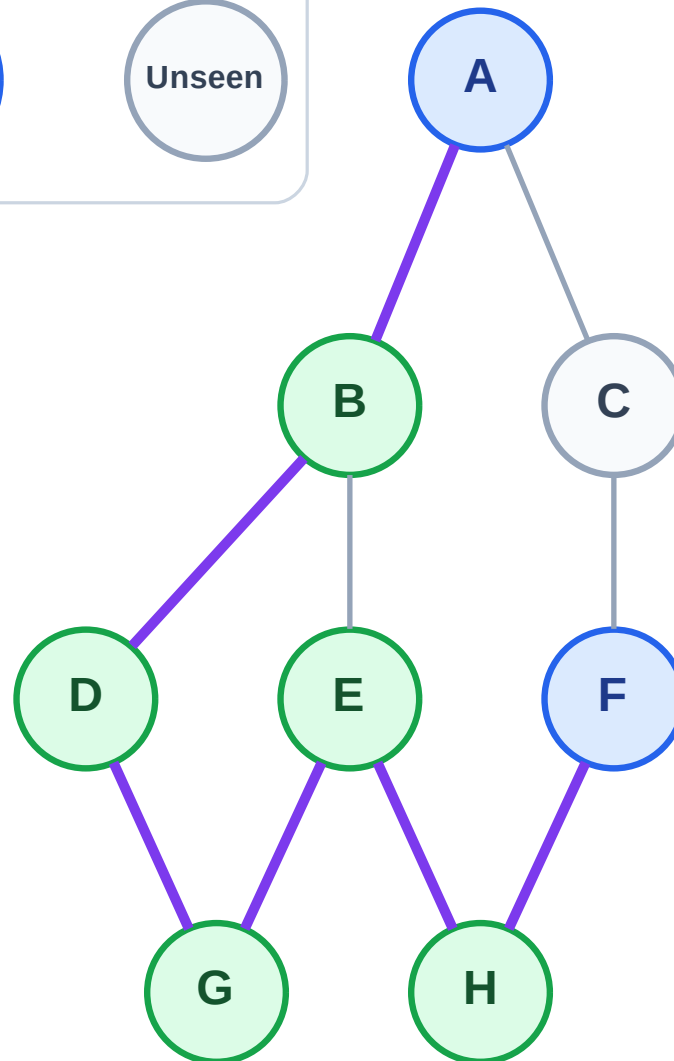
Step 11: Discover F from H

Legend



Queue: A → F

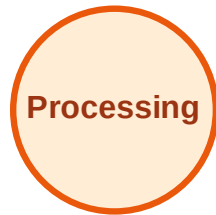
Visited: B, D, E, G, H



BFS Traversal

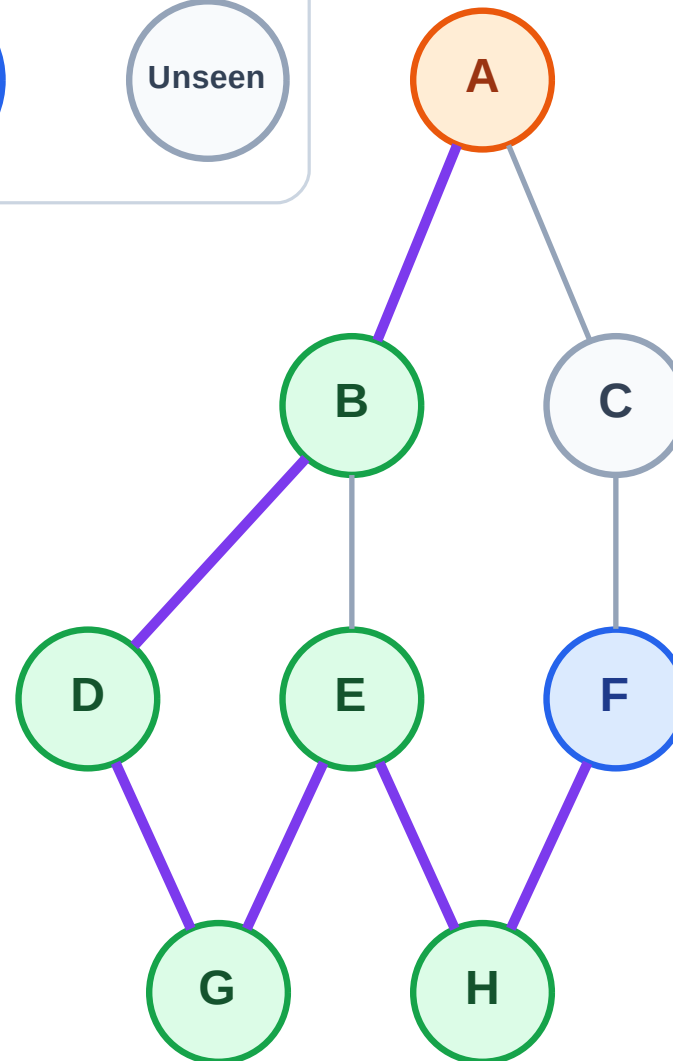
Step 12: Process node A

Legend



Queue: F

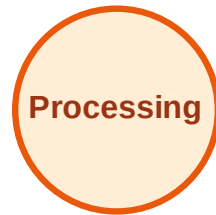
Visited: B, D, E, G, H



BFS Traversal

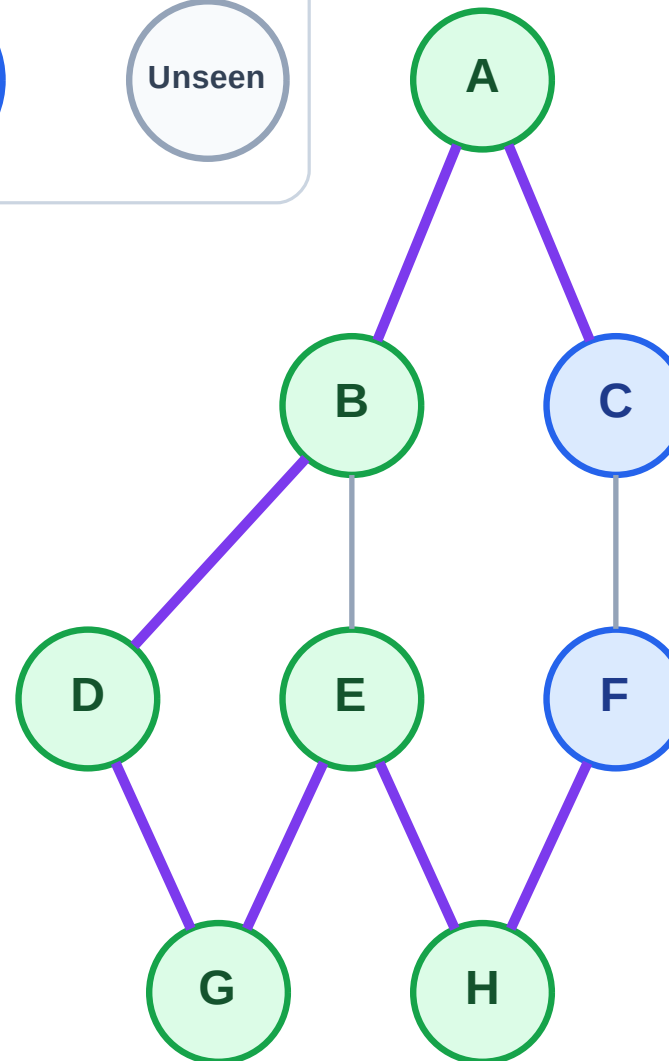
Step 13: Discover C from A

Legend



Queue: F → C

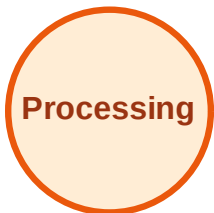
Visited: A, B, D, E, G, H



BFS Traversal

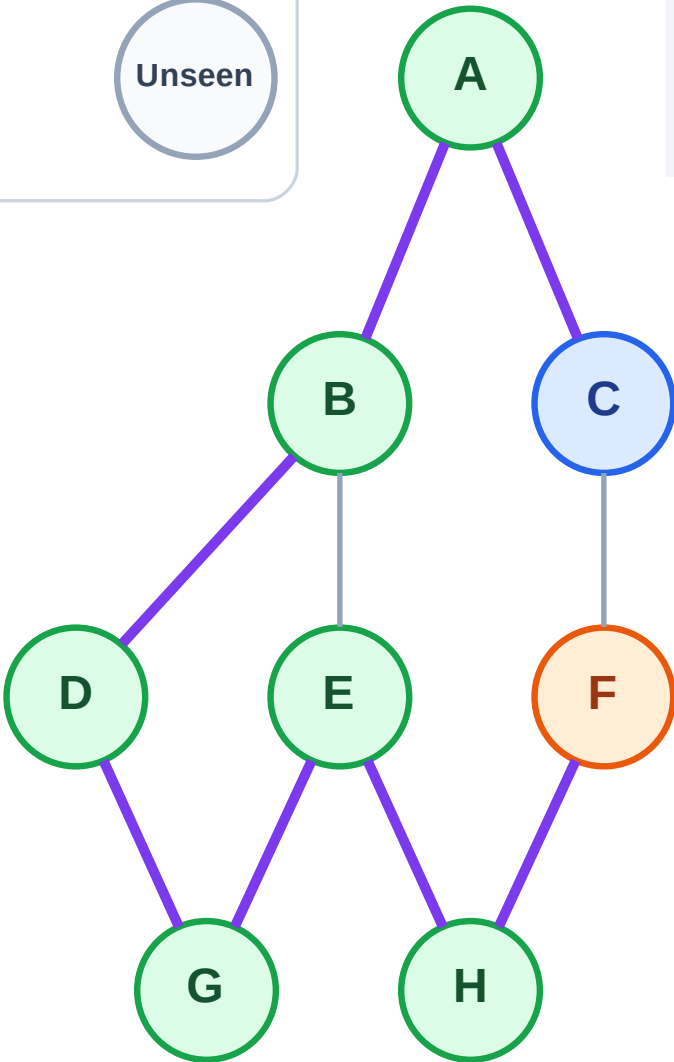
Step 14: Process node F

Legend



Queue: C

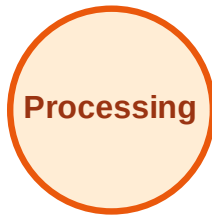
Visited: A, B, D, E, G, H



BFS Traversal

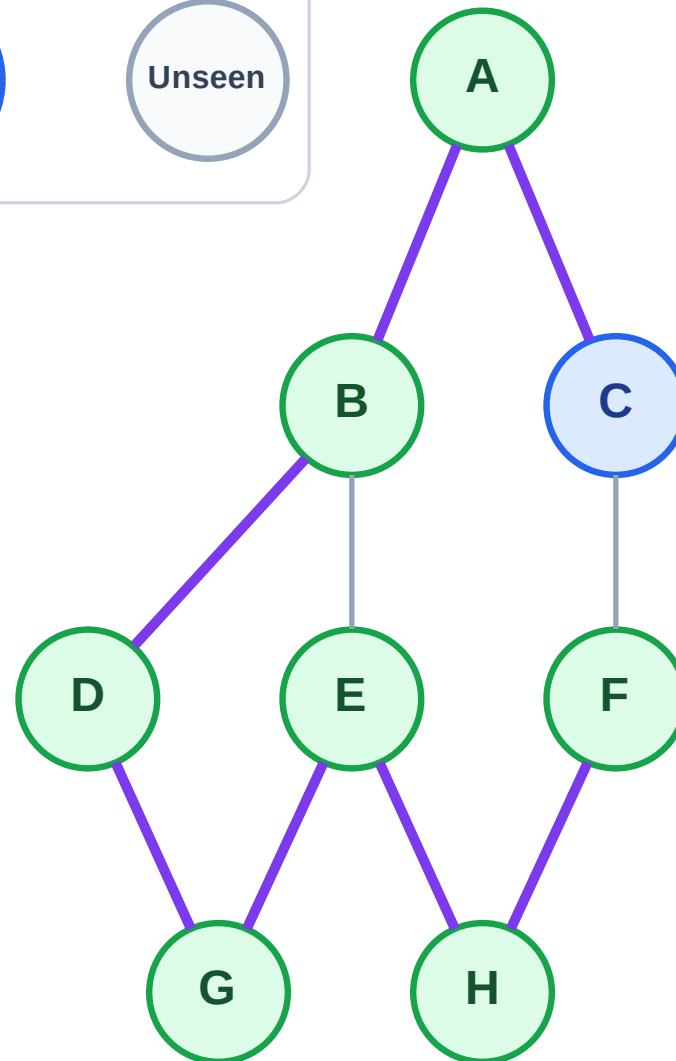
Step 15: No new neighbors from F

Legend



Queue: C

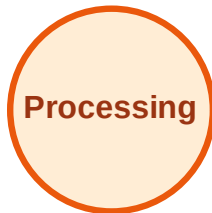
Visited: A, B, D, E, F, G, H



BFS Traversal

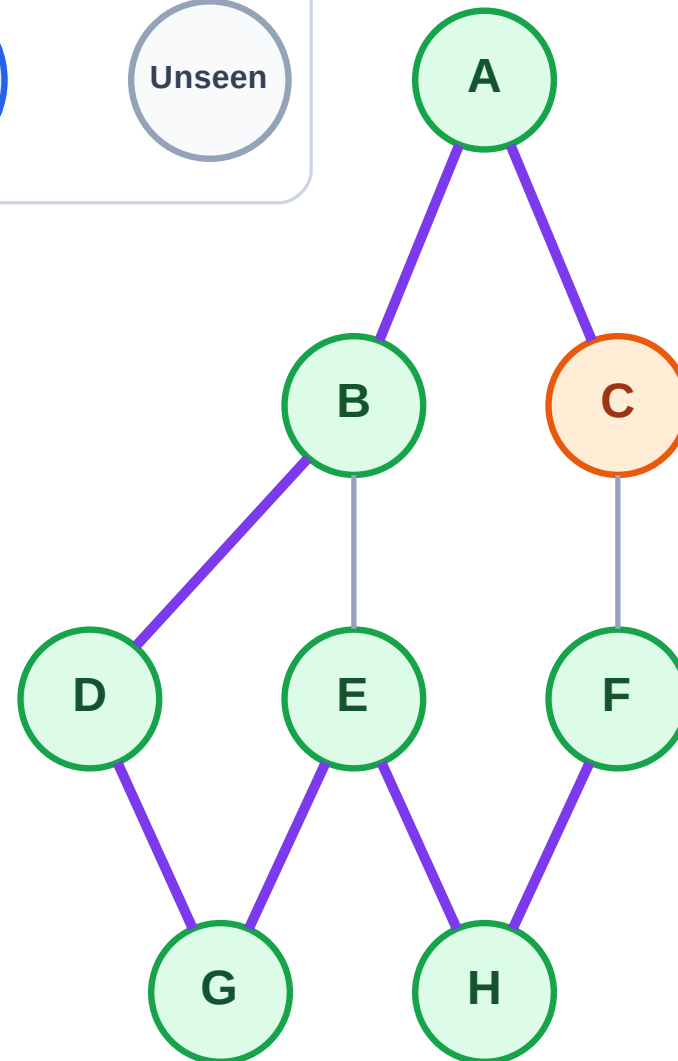
Step 16: Process node C

Legend



Queue: (empty)

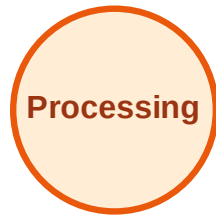
Visited: A, B, D, E, F, G, H



BFS Traversal

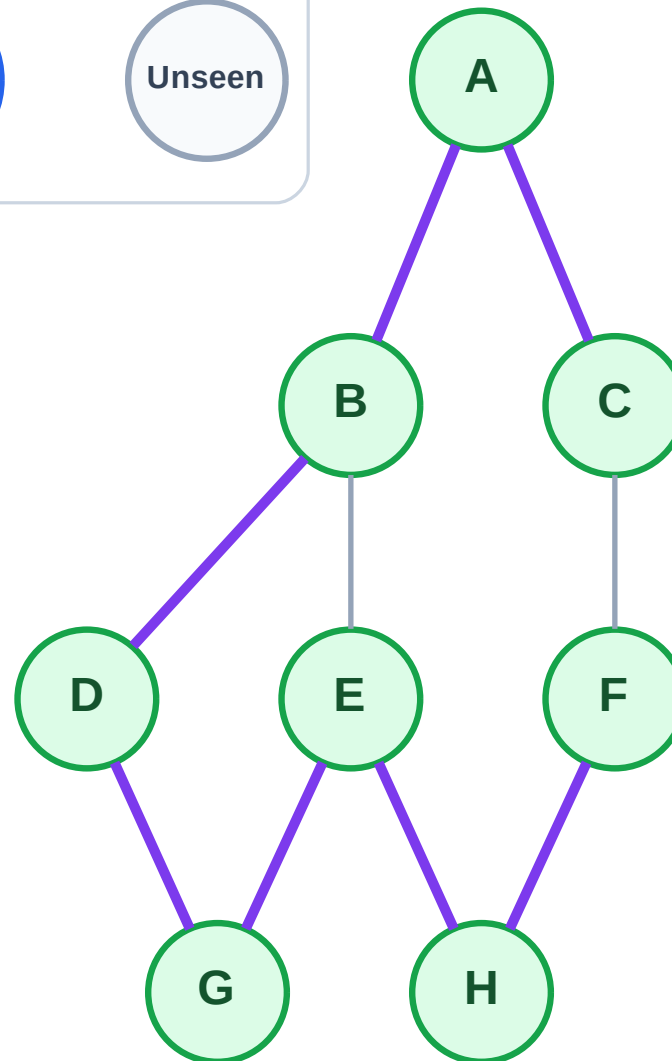
Step 17: No new neighbors from C

Legend



Queue: (empty)

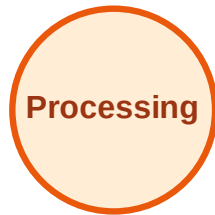
Visited: A, B, C, D, E, F, G, H



DFS Traversal

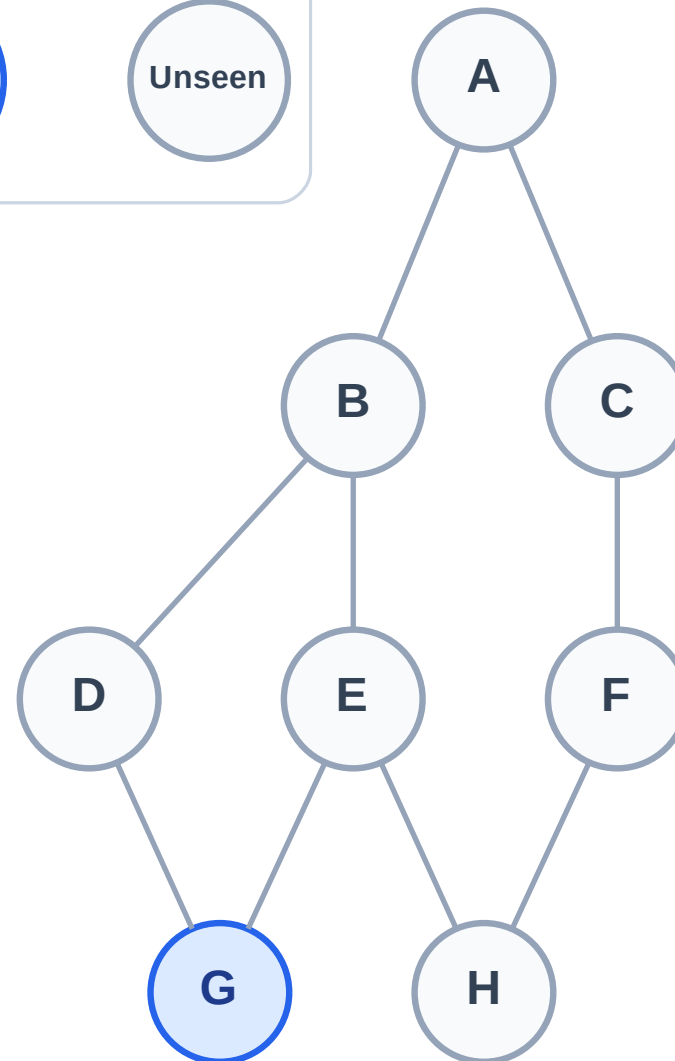
Step 1: Start at G

Legend



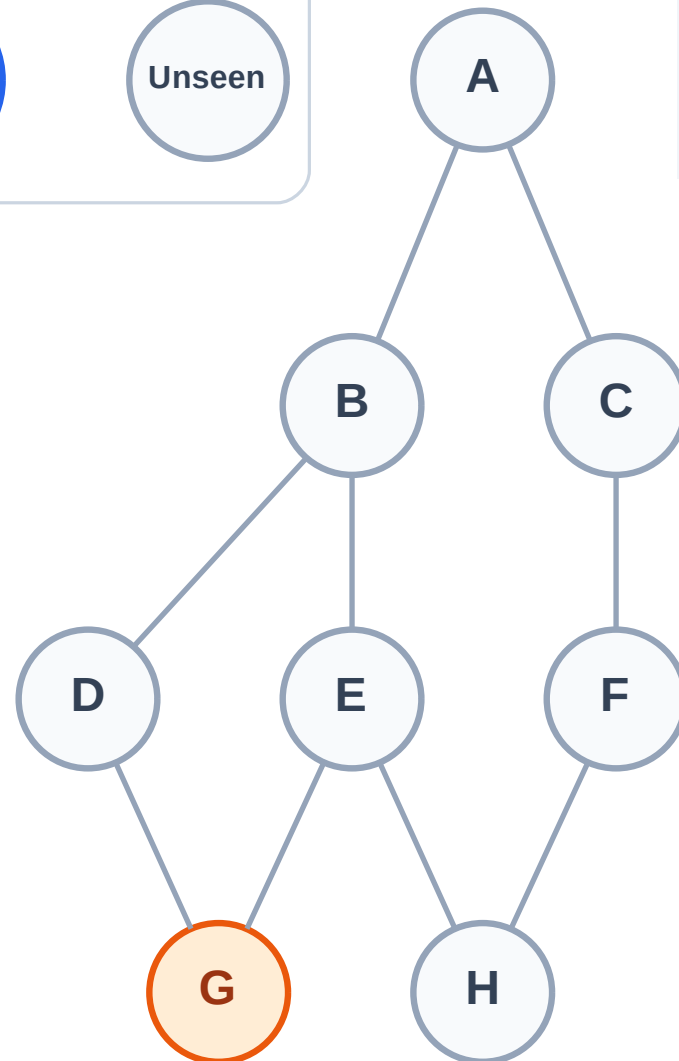
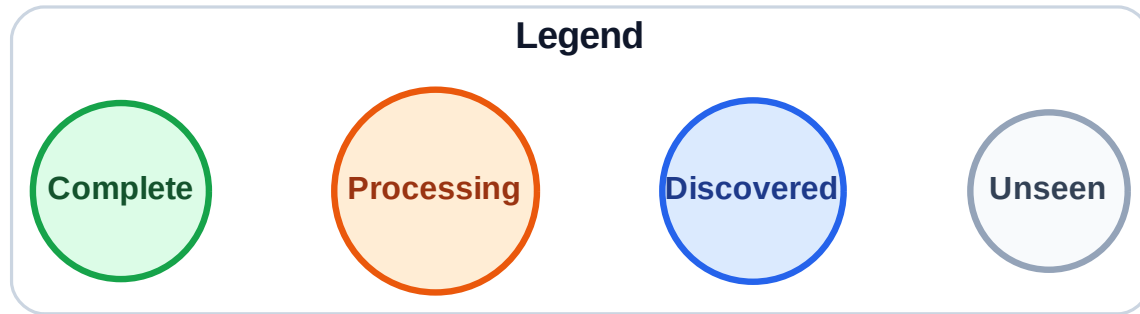
Stack (top at right): G

Visited: (none)



Step 2: Process node G

Step 2: Process node G



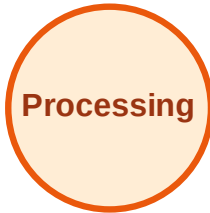
Stack (top at right): (empty)

Visited: (none)

DFS Traversal

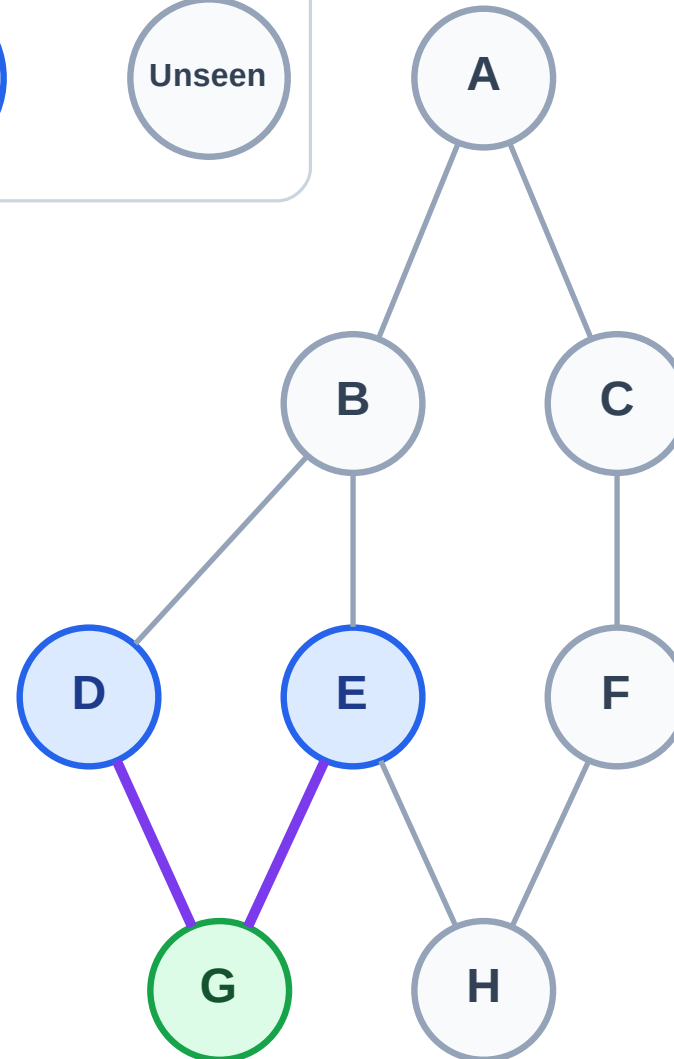
Step 3: Discover E, D from G

Legend



Stack (top at right): E | D

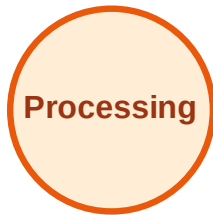
Visited: G



DFS Traversal

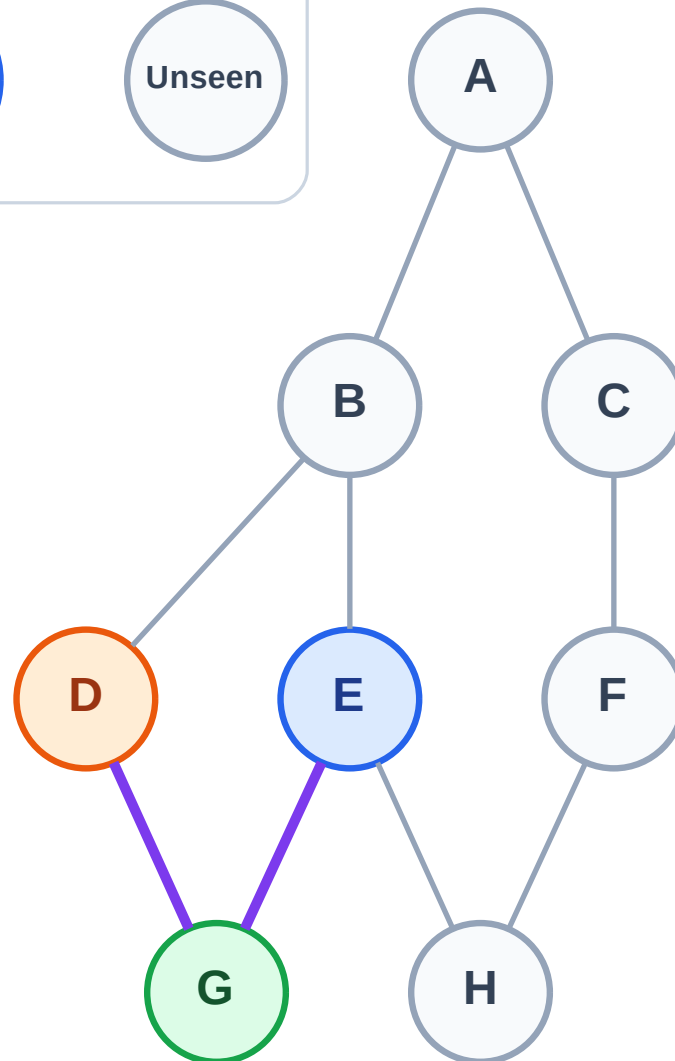
Step 4: Process node D

Legend



Stack (top at right): E

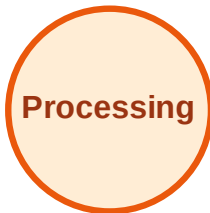
Visited: G



DFS Traversal

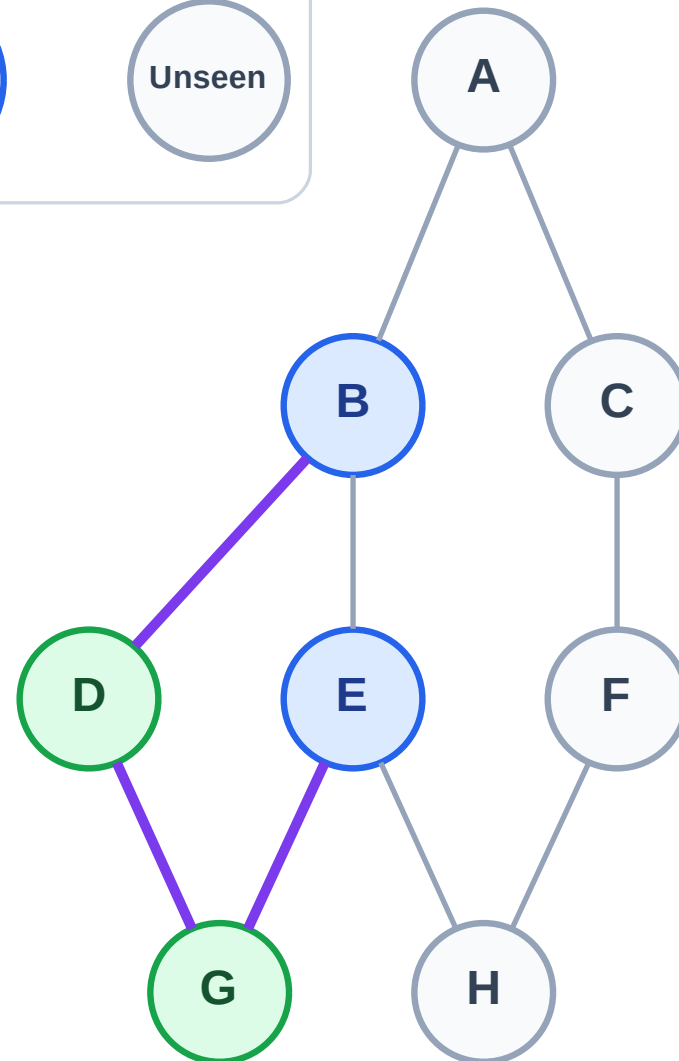
Step 5: Discover B from D

Legend



Stack (top at right): E | B

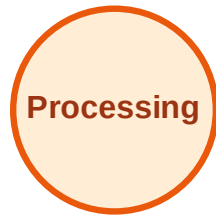
Visited: D, G



DFS Traversal

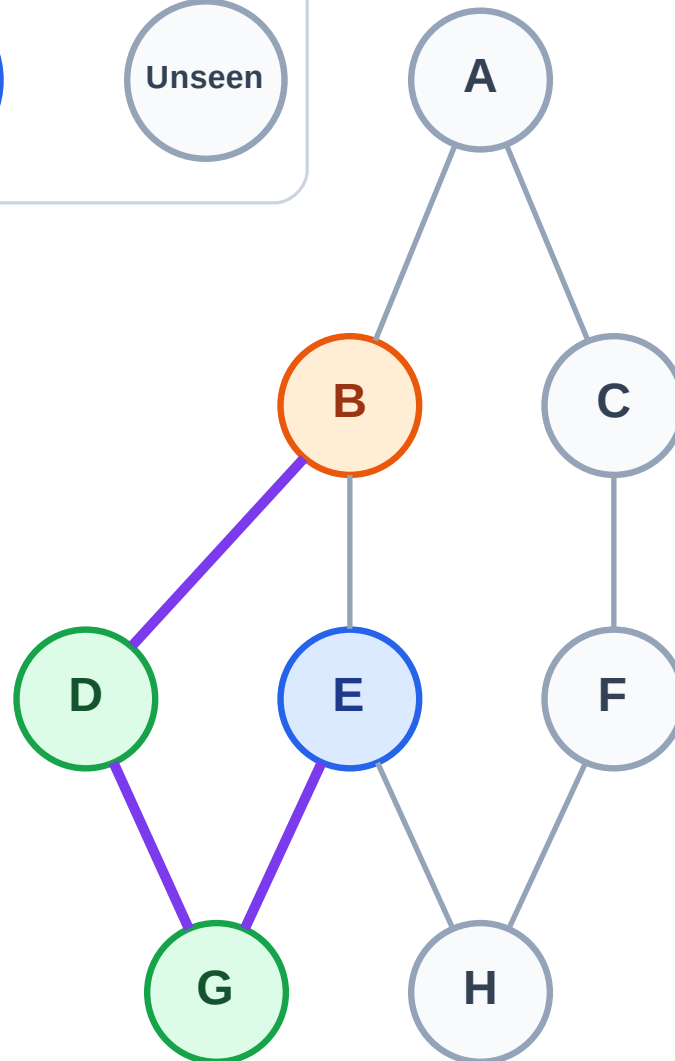
Step 6: Process node B

Legend



Stack (top at right): E

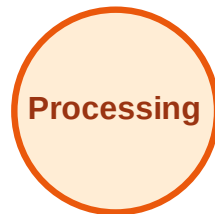
Visited: D, G



DFS Traversal

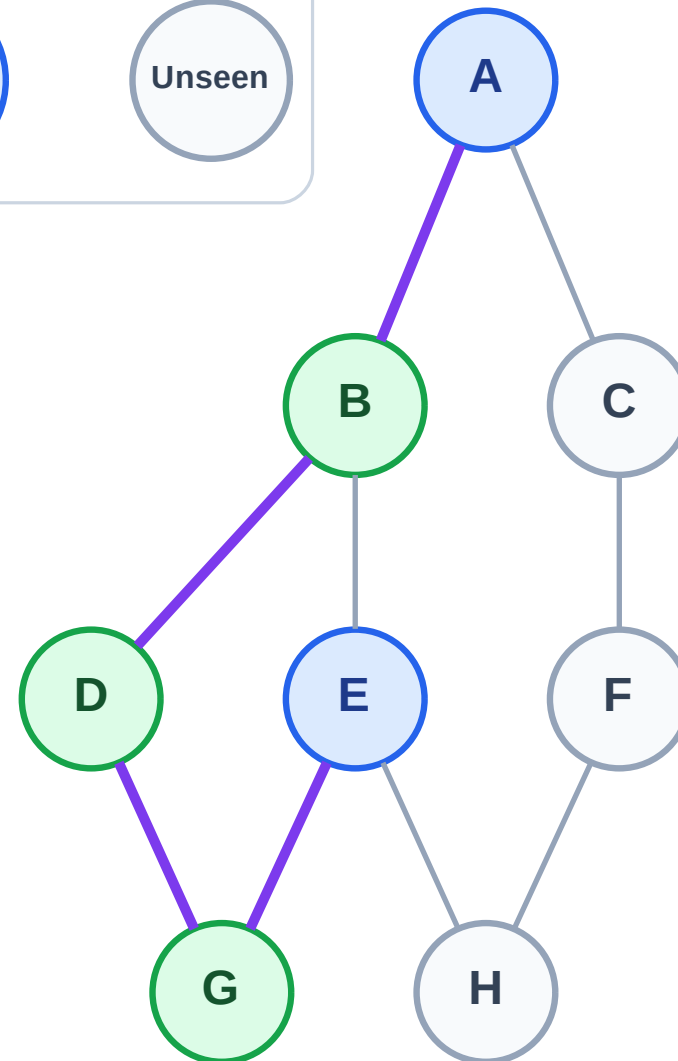
Step 7: Discover A from B

Legend



Stack (top at right): E | A

Visited: B, D, G



DFS Traversal

Step 8: Process node A

Legend

Complete

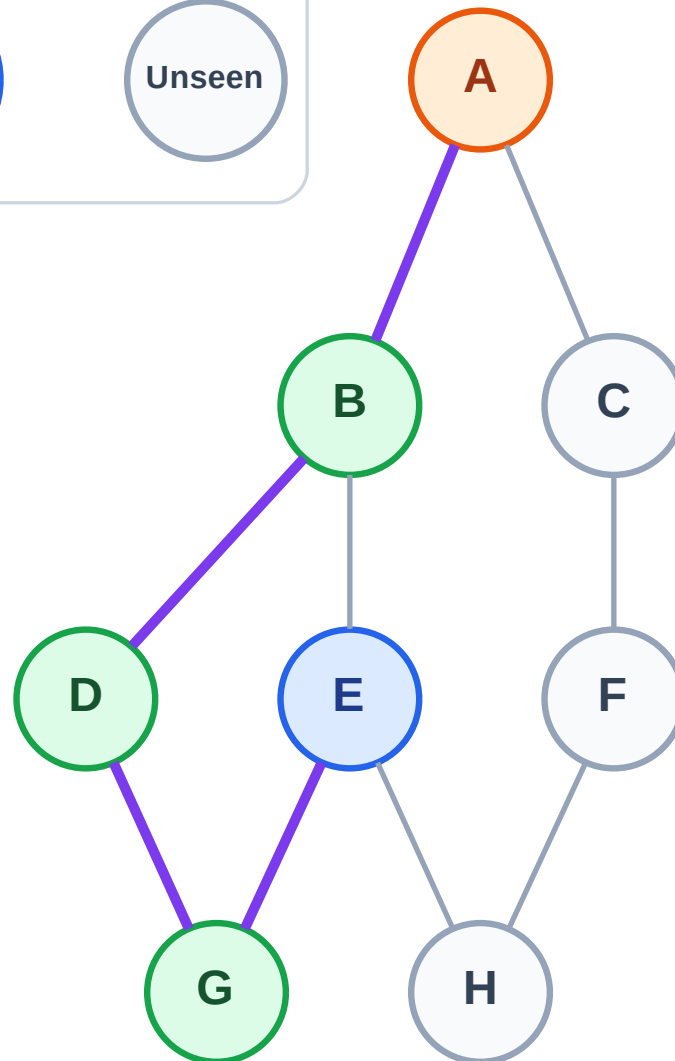
Processing

Discovered

Unseen

Stack (top at right): E

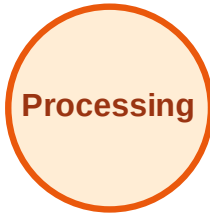
Visited: B, D, G



DFS Traversal

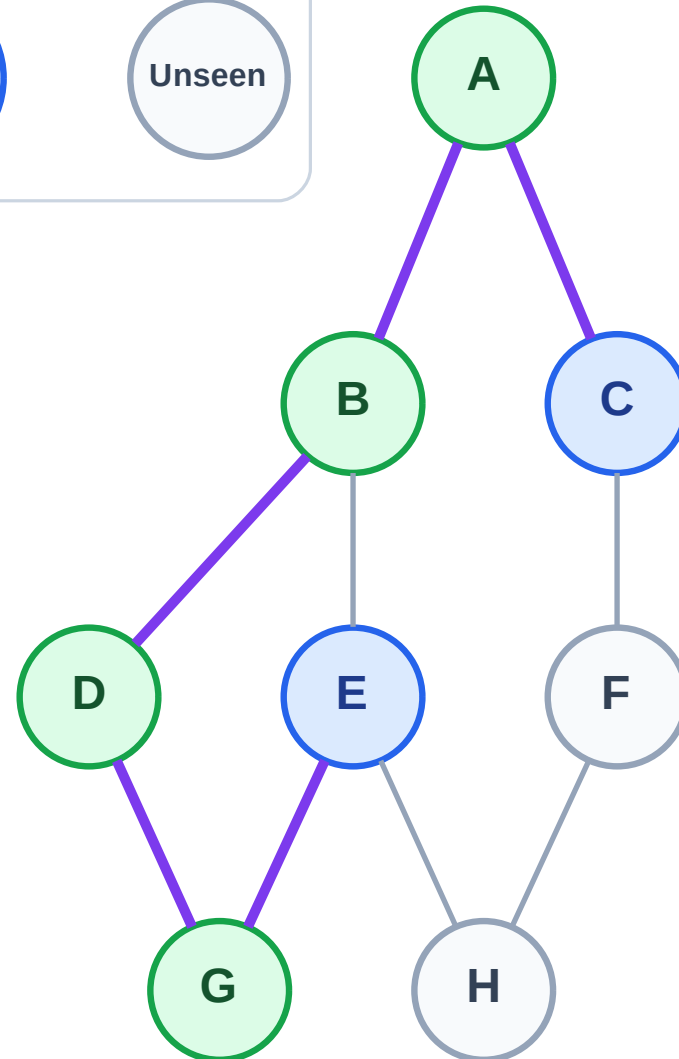
Step 9: Discover C from A

Legend



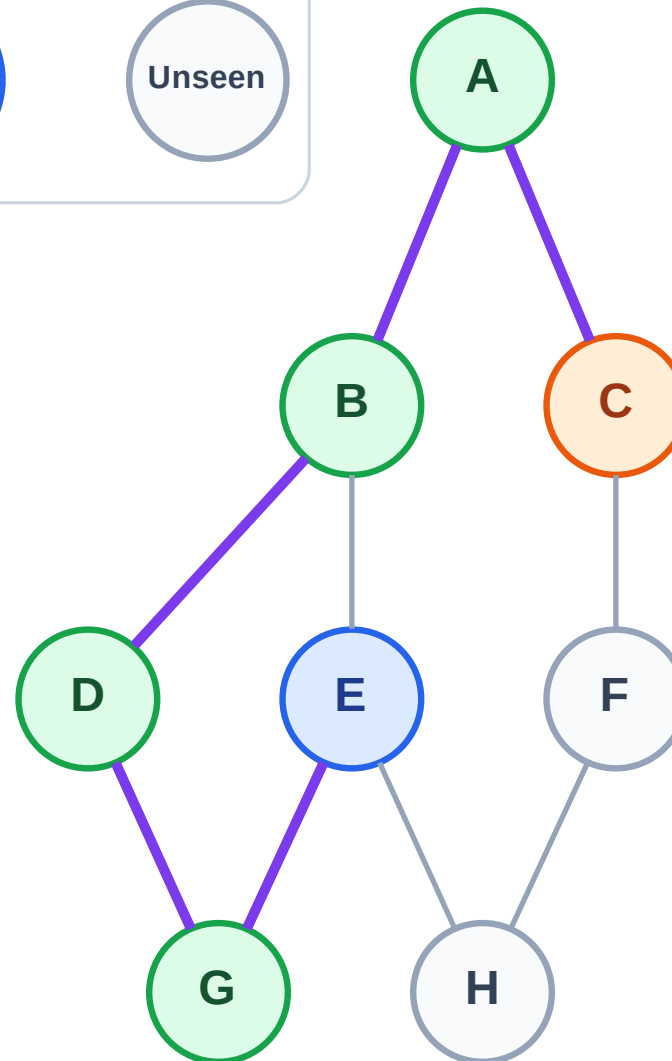
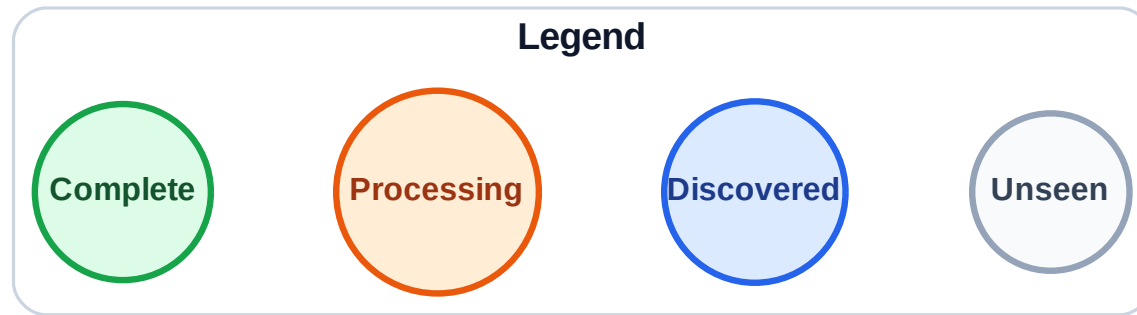
Stack (top at right): E | C

Visited: A, B, D, G



DFS Traversal

Step 10: Process node C

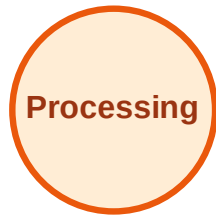


Stack (top at right): E
Visited: A, B, D, G

DFS Traversal

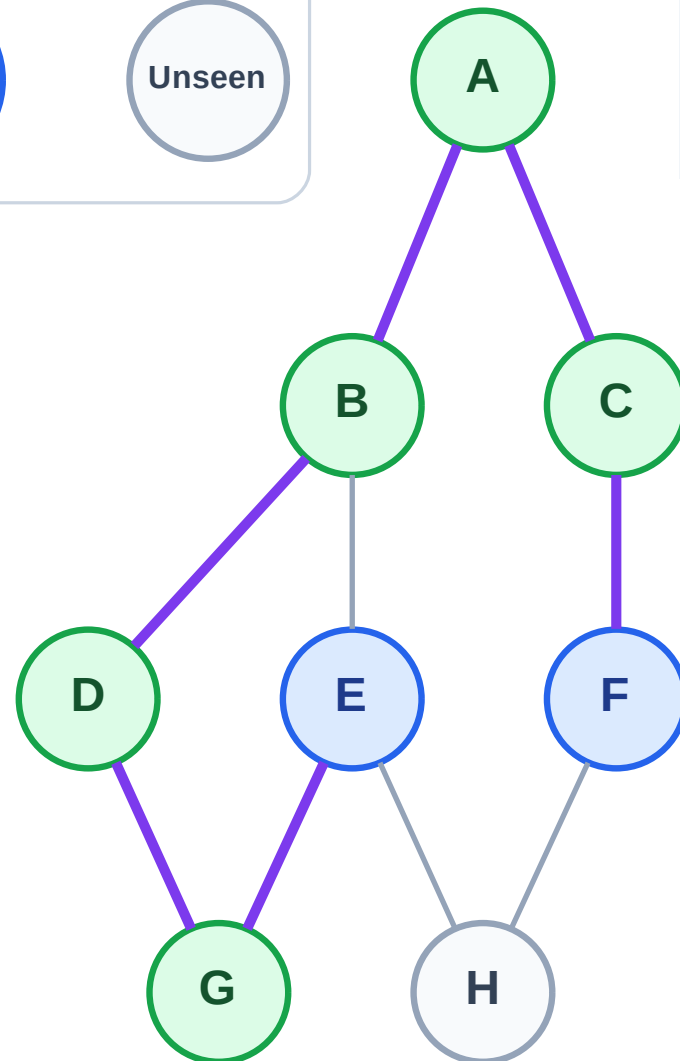
Step 11: Discover F from C

Legend



Stack (top at right): E | F

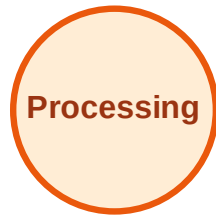
Visited: A, B, C, D, G



DFS Traversal

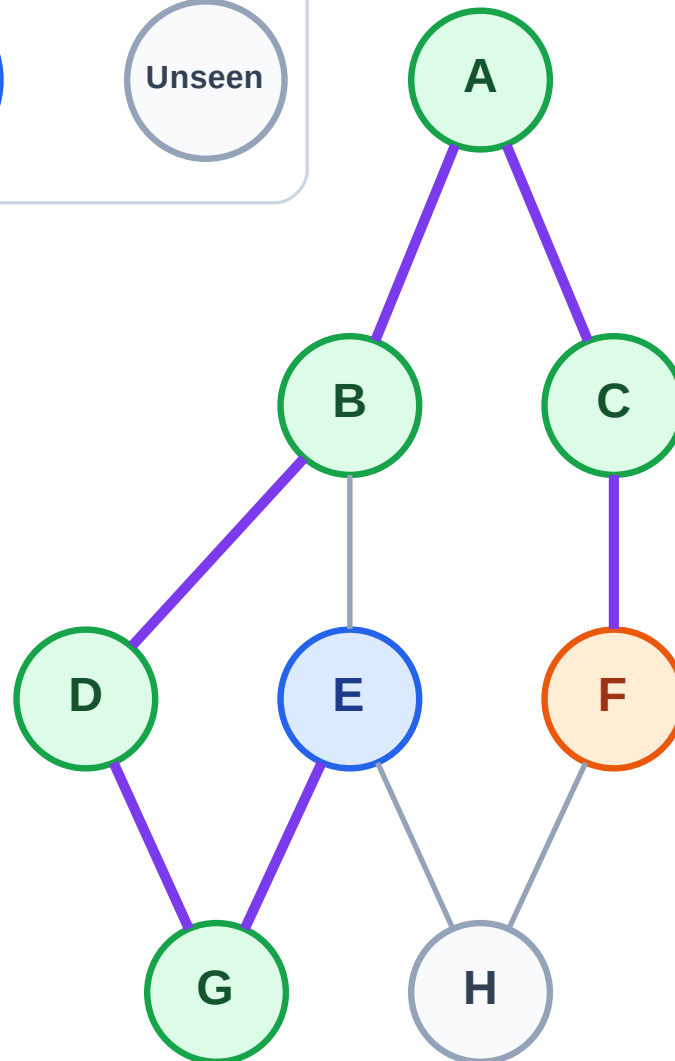
Step 12: Process node F

Legend



Stack (top at right): E

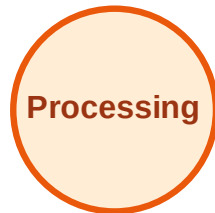
Visited: A, B, C, D, G



DFS Traversal

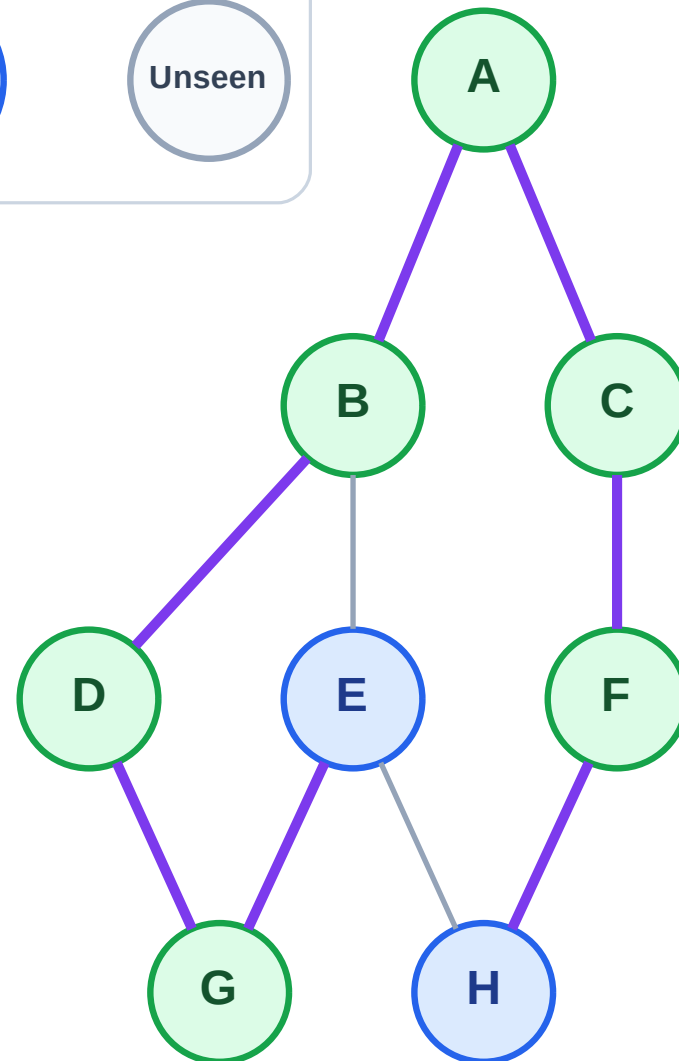
Step 13: Discover H from F

Legend



Stack (top at right): E | H

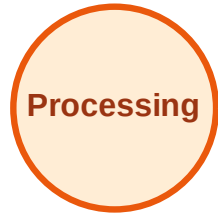
Visited: A, B, C, D, F, G



DFS Traversal

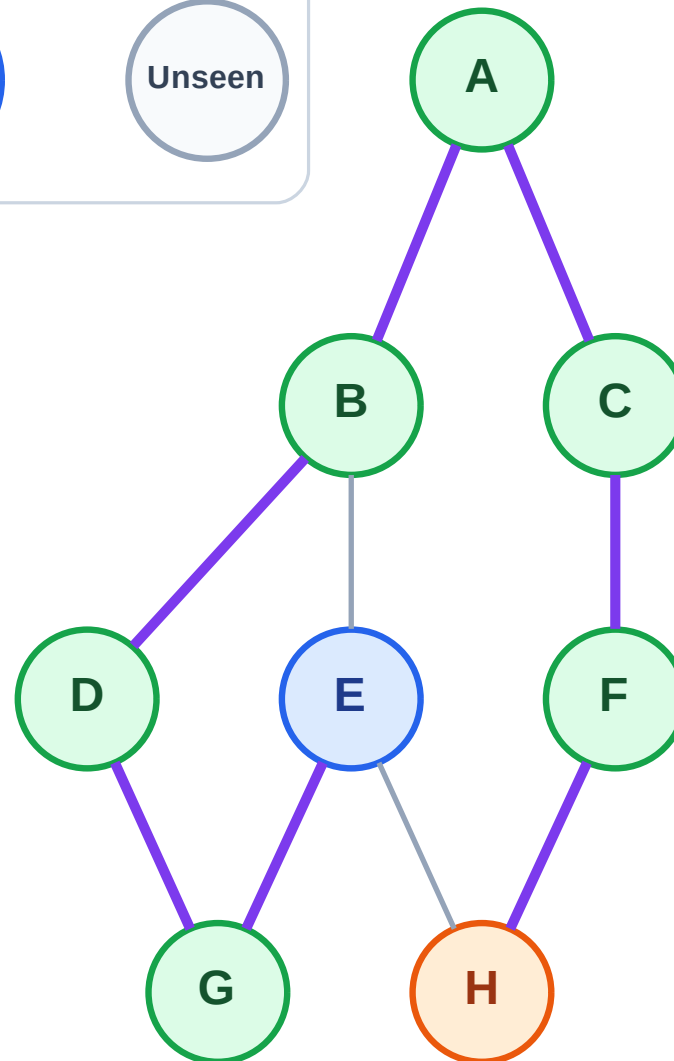
Step 14: Process node H

Legend



Stack (top at right): E

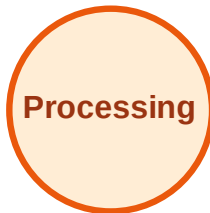
Visited: A, B, C, D, F, G



DFS Traversal

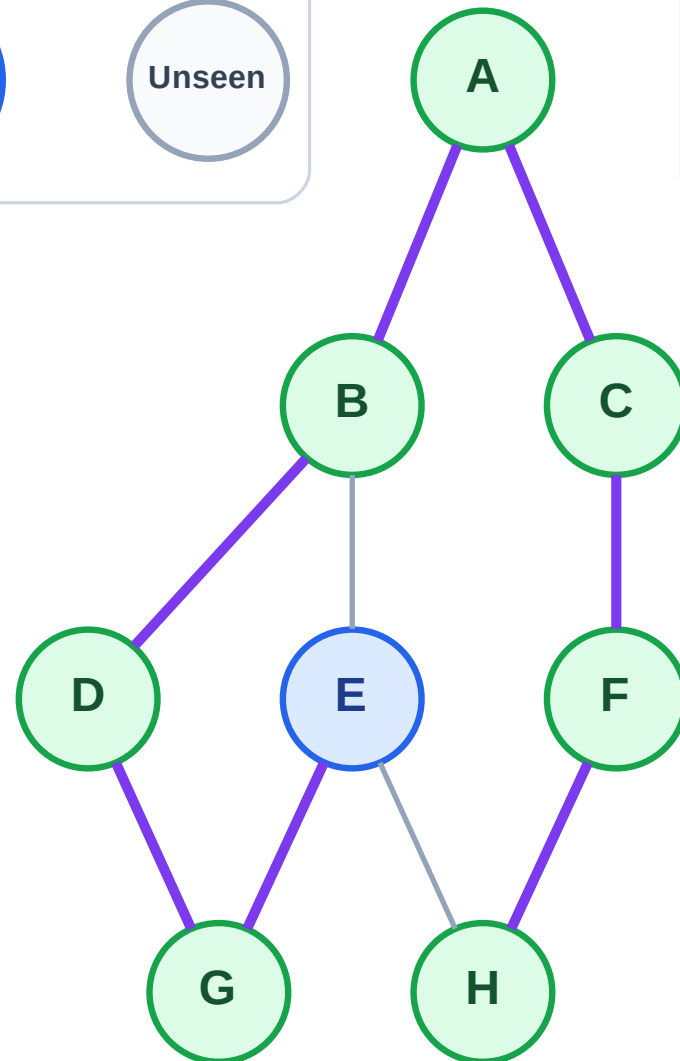
Step 15: No new neighbors from H

Legend



Stack (top at right): E

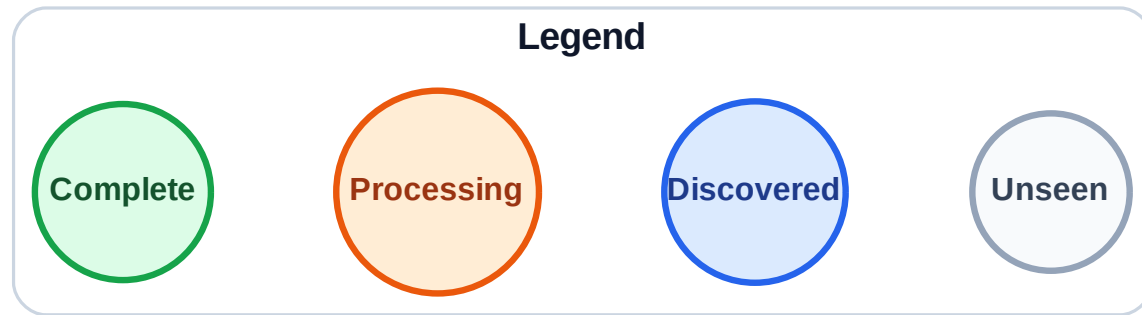
Visited: A, B, C, D, F, G, H



DFS Traversal

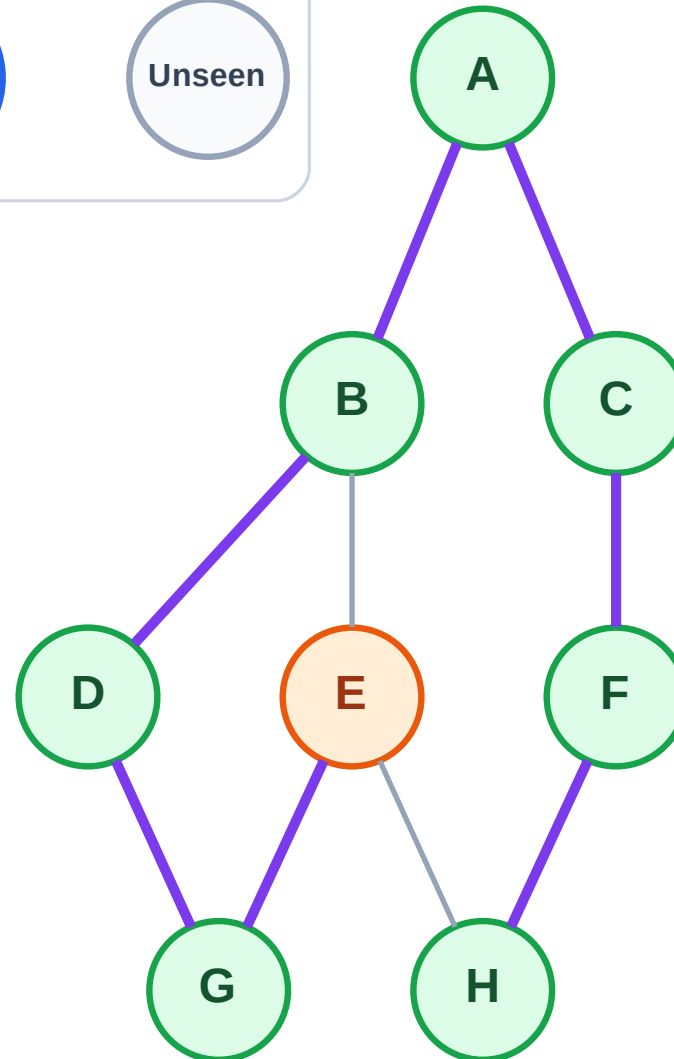
Step 16: Process node E

Step 16: Process node E



Stack (top at right): (empty)

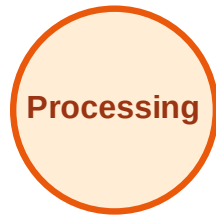
Visited: A, B, C, D, F, G, H



DFS Traversal

Step 17: No new neighbors from E

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H

